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ORIGINAL ARTICLE

Outpatient erbium:YAG (2940 nm) laser treatment for snoring: a prospective study on 40 patients

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Abstract

Snoring is a sleep phenomenon due to the partial upper airway obstruction during sleep which causes vibration of the tissues of the rhino-oro-hypopharynx and less frequently the larynx. This study evaluated the use and effectiveness of the erbium:YAG 2940-nm laser as an adjunctive in providing treatment for patients suffering from chronic snoring-related sleep disorders. A prospective study of 40 consecutive patients with snoring and sleep disorders was performed, assessing data before and after three Er:YAG laser treatment sessions. During laser treatment, the pain was almost absent. There were no side effects, except a very mild sore throat in 1 out of 40 patients. The patient's evaluation of satisfaction of the results obtained after the treatments showed that 85% of cases were very satisfied, 5 patients (12.5%) reported being fairly satisfied with the treatment and only 1 subject (2.5%) was not satisfied. Mallampati, Friedman Tongue Position, and degree of O (oropharynx) at nose oropharynx hypopharynx and larynx classification were significantly decreased after the laser sessions. The decrease of Epworth Sleepiness Scale and Visual Analogue Scale for loudness of snoring, waking up during sleep because of snoring, dry mouth on waking, and choking was all statistically significant. The incidence of dreaming during the night also raised significantly; 30/40 (75%) of cases perceived less tightness in their throat and better breathing after treatment. These results were stable at 20 months follow-up (14–24 q) in 72% of cases. Nonsurgical and non-invasive Er:YAG laser treatment demonstrated to be a valid procedure in reducing the loudness of snoring.

Keywords Laser therapy · Sleep disorder · Obstructive sleep apnea syndrome · Snoring

Introduction

Snoring is a phenomenon that is usually considered innocuous as it can only harm the bed partner of the snorer. Snoring can lead to crisis between couples and, according to the Nypost (Nypost, 9 January 2007 Dr. Rock Positano) in the USA, it is considered the first medical cause of divorce. Additionally, snoring may be very harmful for the patient because it can hide or precede an obstructive sleep apnea syndrome (OSAS) which is associated with hypertension, metabolic syndrome, diabetes, arrhythmias, coronary artery disease, heart failure, pulmonary hypertension, stroke, and neurocognitive and mood disorders [1, 2] as well as a major risk of car accidents through excessive sleepiness with mortality rates double those of other causes.

Snoring may hamper the quality of rest even if OSAS is not present, precipitating awakening because of its loudness. It occurs when a partial obstruction of the upper airways (rhino-oro-hypopharynx and in a minority of cases of the larynx) is present. The negative pressure provoked by the thoraco-abdominal effort to breathe causes a vibration which determines the loudness of snoring. A nasal obstruction can accentuate the negative pressure enhancing snoring itself [3]. The

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key structure in snoring is the soft palate; it may vibrate through negative pressure that develops at this site and its structural length defines the severity of snoring or the appearance of apneas. Additionally, hypertrophic tonsils or an enlarged base of the tongue (for example in people who underwent tonsillectomy) can predispose snoring as well as, in a minority of cases, a "floppy" epiglottis [4]. The major source of vibration causing snoring has been observed in the soft palate and pharyngeal lateral walls while the epiglottis and tongue base usually vibrate slightly [5]. If the obstruction becomes complete, snoring begins to be accompanied by apneas, with a variable degree of severity (OSAS).

To date, simple snoring has been treated with behavioral changes, pillar implant procedure, injection snoreplasty, various other surgical procedures, or with MAD (mandibular advancement device). Pillar implants are designed to work by stiffening the soft palate through the positioning of three tiny woven implants. Submucosal thickening is achieved by creating fibrotic capsules around the implants. Its aim is to decrease snoring and vibration of the soft palate. Postoperative pain usually lasts up to 3 days. This irreversible procedure is an efficient, reliable method in the long term but it could cause complaints such as dysphagia, foreign body sensation, and mouth dryness [6]. Injection snoreplasty uses the principle of palatal stiffening by creating scar tissue in the central part of the soft palate after injecting various sclerosing agents. The sclerosant agents cause palatal ulceration and sloughing, resulting in scar tissue, which is the aim of the procedure; this procedure may cause several days of postoperative pain [7]. Radiofrequency may also be used to cause stiffening of the soft palate by obtaining scar tissue [8] and has postoperative healing similar to injection snoreplasty, with swelling and at least 1 week of pain. There are many types of surgical treatments for snoring; among which, the best known are uvulopalatopharyngoplasty (UPPP), laser-assisted uvulopalatoplasty (LAUP) with an ablative laser, and radiofrequency tissue volume reduction (RFTVR). UPPP is performed under general anesthesia and patients are hospitalized at least for the first night after surgery. UPPP may have some significant complications ranging from respiratory complications, re-intubation, and pneumonia to cardiovascular complications, hemorrhage, and even death. LAUP and RFTVR are considered less invasive as they are performed under local anesthesia, but they have prolonged postoperative pain and many potential side effects, such as problems with smell and taste, pharyngeal dryness, globus sensation, vocal change, and pharyngonasal reflux [9]. All these surgical treatments are invasive, may have complications, and have low success rates and a quite significant number of relapses [10]. Nowadays, patients tend less to accept invasive surgical procedures and often decline to wear an oral appliance or discontinue such therapy in many cases (up to 55% of cases) [11]. Moreover, in the experience of the authors, there are some patients who refuse to wear a continuous positive airway pressure (c-PAP) device, even in cases of severe OSAS and when they are well informed about its risks [12]. Recently, a novel and preliminary outpatient Er:YAG laser treatment has been shown to be effective in reducing snoring and achieved a 65% satisfaction rate as well as improving the quality of sleep and breathing [13]. Based on these considerations, in our work, we wanted to see if we could raise the percentage of satisfaction of patients using a different approach based on the selection of the patient with a previous ear, nose, and throat (ENT) visit with Muller test and treating also the base of the tongue. In this study also, further parameters of sleep-related disorders were analyzed pre- and post-treatment.

Material and methods

The cohort for the present study consisted of 40 patients (29 male and 11 female, average age 53 years), who presented for snoring that caused relationship problems with bed partners, to the private practice of one of the authors (IFS). Treatment was provided using the erbium yttrium aluminum garnet laser (Er:YAG 2940 nm; LightWalker, Fotona, Slovenia). Five patients (12.5%) of these 40 presented for snoring and OSAS and they refused any other kind of treatment, including c-PAP, oral appliance (MAD), or surgery; preference was given for a non-invasive multi-step attempt from non-invasive to more invasive procedures to improve their apneas. The exclusion criteria were pediatric patients, pregnancy, and central (neuropathic) apneas.

All the following data were assessed before and after the treatment, in order to quantify the efficacy of the procedure or the better outcome of certain features for the patient.

During the pre-treatment, all patients underwent an ear, nose, and throat (ENT) examination visit with a modified Muller maneuver (FNMM) [14] with video-fibroscopy. The naso-pharyngoscope (Xion, Germany), connected to a high-resolution video system (Karl Storz Endovision TRICAM, Tuttlingen, Germany), was introduced through the nose to assess the anatomy of the upper airway. The FNMM was performed in the supine position without any pillow and the Muller test with NOHL (nose oropharynx hypopharynx and larynx) classification [15] was established, in order to evaluate the major site of obstruction and exclude cases with non-suitable situations, i.e., floppy epiglottis causing laryngeal obstruction that cannot be reached with this outpatient laser treatment. The presence of laryngo-pharyngeal reflux (LPR) was evaluated with video-fibroscopy (Karl Storz, Germany), using Reflux Symptom Index (RSI) and Reflux Finding Score (RFS) [16]. Before treatment, the patients were divided into four classes according to Mallampati classification (class 1—full visibility of tonsils, uvula, and soft palate; class 2—visibility of hard and soft palate, upper portion of tonsils, and uvula; class 3—soft and hard palate and base of the uvula are visible; class 4—only hard palate visible) [17] and into 4 classes according to the FTP (Friedman Tongue Position) (Fig. 1) [17]. A body mass index (BMI) from 25 to 29.9 was considered overweight and a BMI of 30 or higher indicated obesity. Daytime sleepiness was evaluated using the Epworth Sleepiness Scale (ESS) [18] ranging from 0—lower daytime sleepiness—to 24—excessive daytime sleepiness. Patients who reported (together with their bed partner) an ESS over 9 were invited to perform a polysomnography to assess the presence of apneas. The same exam was also recommended in case of other suspect symptoms such as choking or the effective presence of apneas that were noted by the bed partner of the snorer. The polysomnography was analyzed by the same referent and using the same polygraph (CareFusion Noxturnal). The exception was for five patients who presented for snoring and OSAS and were examined in another medical practice.

The following parameters were scored by each patient and his/her bed partner before and after the treatment in order to quantify the efficacy of the procedure using a VAS (Visual Analogue Scale) from 1 to 10 pre- and post-laser treatment: the rating of snoring intensity (0—no snoring—up to 10—extreme snoring causing the bed partner to leave the bedroom) [8], difficulty to wake up at mornings, waking up during sleep because of snoring, dry mouth in the morning, subjective absence of dreaming during the night.

The treatment was performed on an outpatient basis.

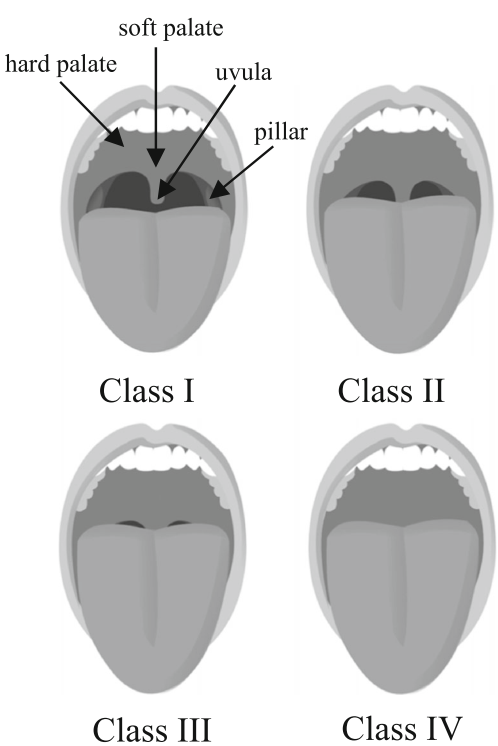

Fig. 1 FTP and Mallampati classification (Jmarchn, 2017, modified) [35]

Informed consent was obtained prior to initiating treatment and all the patients with OSAS were first informed to perform other kind of evidence-based treatments such as wearing c-PAP in moderate or severe OSAS, wearing MAD, or surgery, but they all refused to perform any other kind of treatment considered invasive or uncomfortable. Five cases already knew they were OSAS patients and wanted to try a mini-invasive treatment as they refused surgery, c-PAP, or MAD and were informed that in case of failure with OSAS, they could eventually perform multi-step treatments with an increasing degree of invasiveness and always controlling the results, step by step with polysomnography.

The patients were all treated with the Er:YAG laser in a non-contact mode with irradiation of at least 7–8 pass of all regions with an overlap covering the entire mucosal surface, using a PS04 handpiece and collimated beam. The parameters were LP mode, 10 Hz, and fluence in the range of 1.6 J/cm^2. The number of delivered treatment pulses per region and per patient depended on the severity of Muller test score, on the anatomy of the person, on the presence or absence of apneas, and on the severity of the symptoms and varied from a minimum of 11,086 to a maximum of 25,689 shots.

Three sessions were performed at 0–15 and 45 days. The areas that were exposed to laser energy were soft palate and uvula and tonsillary regions, including the anterior and posterior pillars and the base of the tongue behind the circumvallate papillae as far as the anatomy of the patient and his compliance allowed (Fig. 2). Gag reflexes were overcome using relaxing breathing techniques, acupuncture point stimulations, or a topical lidocaine spray. The mechanism of action of the erbium:YAG laser is a photo-thermal effect, which causes shrinkage of the collagen fibers in the treated oral mucosa and initiates, through heat shock protein (HSP) action, a neo-collagenesis [19–22].

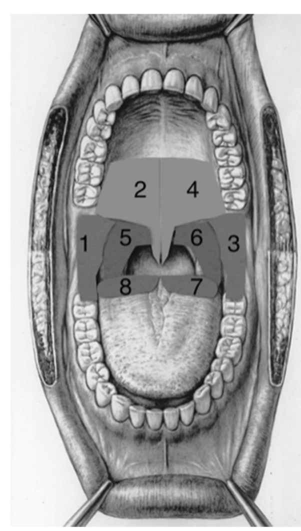

Fig. 2 Treated regions (Winter, 2017, modified) [36]

During the treatment, the pain response was measured on a Visual Analogue Scale (VAS) (from 0 = no pain to 5 = very severe pain). At the end of the treatment, the patient valued his/her immediate perception of a "wider throat" from a range from 0 to 5 and patients were asked about adverse effects.

Statistical analysis

Continuous variables are summarized as median (1st–3rd quartiles) and categorical variables as frequencies and percentages. Comparison of quantitative variables between pre- and post-treatment was made by Wilcoxon matched-pairs signed-ranks test, and comparisons of frequencies were made by McNemar test.

Values of $p < 0.05$ were considered statistically significant. All analyses were performed using SAS version 9.3 (SAS Institute, Cary, NC).

Results

All patients completed all three sessions of the therapy. In 17 of 40 patients (42.5%), OSAS was first diagnosed performing a polysomnography for an altered ESS (Epworth Sleepiness Scale) or other suspect symptoms (choking, apneas noted by the partner). There were 22 OSAS patients; 4 patients had an initial OSAS (Apnea-Hypopnea Index (AHI) 5–15), 10 were moderate (AHI 15–30), and 8 severe OSAS (AHI > 30); 13 of these reported frequent choking during sleep.

Using the Mallampati classification, 20 of them (50%) were classified as class 4, 10 patients as class 3 (25%), 9 patients as class 2 (22.5%), and only 1 patient as class 1 (2.5%). Twenty subjects (50%) had a BMI lower than 25, 11 subjects (27.5%) were overweight, and 9 subjects (22.5%) were obese. Demographic and clinical characteristics of the enrolled patients were reported in Table 1.

Table 1 Demographic and clinical characteristics of the 40 patients enrolled at baseline

Median age (years)	51 (43–63.25)
BMI	24.9 (22.9–29.3)
Drug allergy	8 (20%)
Smoke	
Non smokers	31 (77.5%)
Current smokers	7 (17.5%)
Ex-smokers	2 (5%)
Comorbidities	15 (37.5%)
Quality of breath	
Not good	21 (52.5%)
Acceptable	9 (22.5%)
Good	10 (25%)
Assumption of concomitant drugs	17 (47.2%)
Septum	
Aligned	8 (20%)
Minor	5 (12.5%)
Moderate	11 (27.5%)
Severe	16 (40%)
Tonsils	
Intravelic	9 (22.5%)
Extravelic < 50%	12 (30%)
Extravelic > 50%	4 (10%)
Tonsillectomy	15 (37.5%)
Soft palate	
Regular	7 (17.5%)
Slackened grade 1	18 (45%)
Slackened grade 2	13 (32.5%)
Slackened grade 3	2 (5%)
Uvula	
Regular	9 (22.5%)
Slightly hypertrophic	16 (40%)
Severely hypertrophic	15 (37.5%)
Reinke's chronic vocal fold edema	4 (10%)
Laryngo-pharyngeal reflux	22 (55%)
Obstructive sleep apnea syndrome	22 (55%)

All patients evaluated the pain during the therapy. After the first session, the median pain was 1 point (0–2), after the second it was 1 (0–1.7), and after the last one it was 1 (0–1) on the 0–5 pain scale. Gag reflexes were overcome using relaxing breathing techniques and acupuncture stimulations for the majority of patients. Only 4/40 patients (10%) received a topical lidocaine spray at the first session. In the successive laser sessions, gag reflexes lowered in all patients and there was no further need to use topical lidocaine. There were no other adverse effects of this laser therapy noted at any of the three sessions except a mild sore throat for a few hours in 1 of 40 (2.5%) treated subjects—these required no analgesic treatment and ceased spontaneously.

Patients were asked regarding their own satisfaction after the three laser sessions. Of the 40 treated patients, 34 (85%) were satisfied after the laser treatment, 5 patients (12.5%) reported being fairly satisfied with the treatment, and only 1 subject (2.5%) was not satisfied. All assessments were reevaluated 1 month after treatment with a final visit.

The Mallampati classification was significantly decreased after the laser treatment ($p = 0.001$), as well as the FTP and "O" (oropharynx) grade of obstruction at the NOHL classification ($p = 0.001$ for both) (Fig. 3).

Of 20 patients classified as class 4 from the Mallampati classification, 12 (60%) were classified as class 3 or 2 after the laser treatment, while 8 (40%) patients did not show any change; of the 23 subjects classified as class 4 from the FTP scale, almost 61% were classified as class 3 or 2 after the treatment.

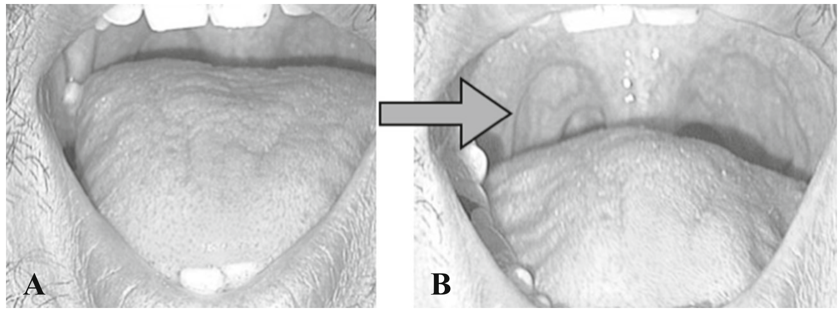

Fig. 3 Improvement of FTP from grade 3 (**a**) to grade 2 (**b**)

After the laser treatment, 30/40 (75%) patients reported to have an immediate sensation of breathing improvement and wider throat.

Apart from each patient's general assessment of snoring reduction, for more detailed result evaluation, the patients were provided with questionnaires assessing various aspects of patient's snoring experience before and after the treatment (Table 2).

Snoring severity significantly improved after the procedure. The median snoring VAS score according to the bed partners was reduced from a median value of 10 to 3 ($p < 0.0001$) which allowed to four couples that slept in separate beds to sleep together again while the daytime sleepiness, assessed with the ESS score, was reduced from 4 to a median value of 2 ($p < 0.0001$).

The quality of the sleep showed a significant increase from an initial median value of 5 to a median value of 10 ($p < 0.0001$), and also, the intensity of the dreaming perceived by the patients significantly increased ($p < 0.0001$). In particular, 15 patients who never reported dreams before the treatment began to dream again from the day after the first session.

Mouth dryness at wake-up time significantly decreased after the laser treatment ($p < 0.0001$), as well as the difficulty in waking in the morning ($p < 0.001$) and the waking up during sleep because of snoring ($p < 0.0001$).

Notably, all but one (12 out of 13 patients) of the patients reporting choking before the treatment reported that this symptom had ceased ($p = 0.001$).

The subgroup analysis according to BMI categories and obstructive sleep apnea syndrome (OSAS) did not show any significant difference in reporting satisfaction of the treatment effect and improvement in breathing and in the variables assessing various aspects of patient's snoring situation.

The change in patient general assessment of snoring reduction, pre- and post-laser treatment, was significant in patients with and without OSAS and in normo-weight and overweight patients.

A subgroup analysis was performed to compare the improvement in snoring reduction in patients with or without tonsillectomy (excluding four patients with extravelic tonsil > 50%).

The change of patient's general assessment of snoring reduction, pre- and post-laser treatment, was significant in patients with and without tonsillectomy and only the improvement in dreaming and the reduction of the difficulty to wake up at mornings were non-significant in patients with tonsillectomy ($p = 0.06$ and $p = 0.13$, respectively).

Table 2 Comparison of patient's general assessment of snoring reduction variables pre- and post-laser treatment

	Pre-treatment $N = 40$	Post-treatment $N = 40$	p value
Friedman Tongue Position			0.001
Class 4	23/39 (59.0)	9/39 (23.1)	
Class 3	11/39 (28.2)	19/39 (48.7)	
Class 2	4/39 (10.3)	8/39 (20.5)	
Class 1	1/39 (2.5)	3/39 (7.7)	
Mallampati			0.001
Class 4	20/38 (52.6)	8/38 (21.1)	
Class 3	8/38 (21.1)	17/38 (44.7)	
Class 2	9/38 (23.7)	7/38 (18.4)	
Class 1	1/38 (2.6)	6/38 (15.8)	
Muller test (oropharyngeal)			0.001
Class 4	23/38 (60.5)	9/38 (23.7)	
Class 3	14/38 (36.8)	23/38 (60.5)	
Class 2	1/38 (2.6)	5/38 (13.2)	
Class 1	0/38 (0.0)	1/38 (2.6)	
Epworth Sleepiness Scale	4 (2–10)	2 (0–5)	< 0.0001
Epworth Sleepiness Scale			0.005
≤ 9	28/39 (71.8)	36/39 (92.3)	
> 9	11/39 (28.2)	3/39 (7.7)	
VAS snoring	10 (8–10)	3 (1–5)	< 0.0001
Restful sleep	5 (0–10)	10 (8–10)	< 0.0001
Difficulty to wake up	0 (0–7.5)	0 (0–0)	< 0.001
Dreaming at night	2 (0–10)	8 (6–10)	< 0.0001
Waking up during sleep	0.5 (0–5)	0 (0–0)	< 0.0001
Dry mouth	6 (0–10)	0 (0–0)	< 0.0001
Choking	12/37 (32.4)	1/37 (2.7)	< 0.001
AHI	16.4 (6.4–27.7) $N = 11$	15.1 (6.4–19) $N = 11$	0.08

Table 3 Differences in various treatments for snoring

	ER:YAG laser	Pillar implants	C-PAP	LAUP	UPPP	RFTVR ablation	Sclerotherapy	Oral appliance
Pain or discomfort	Low /none	Low	Low/medium	High	High	Low/medium	Low/medium	Low/medium
Potential side effect/most reported complication	None	Partial extrusion (< 1%)	Nocturnal awakenings (46%), nasal congestion and dryness (44%)	Transient VPi (27%)	Transient VPi (22%)	Mucosal ulceration and breakdown	Mucosal ulceration and breakdown	TMJ problems
Sedation	None	Local	None	Local/general	General	Local	Local	None
Recovery time	None	24 h or less	N/A	7 days	Up to 2 weeks	24 h or less	24 h or less	None
Snoring	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
OSAS	N/A	N/A	Yes	Yes	Yes	No	No	Yes
Physician time	Medium	Low	Low	High	High	Medium	Medium	Medium

VPi velopharyngeal inadequacy, *LAUP* laser-assisted uvulo-plasty, *UPPP* uvulopalatopharyngoplasty, *RFTVR* radiofrequency tissue volume reduction, *TMJ* temporo-mandibular joint

Notably, in the four subjects with extravelic tonsil > 50%, the treatment did not seem to be effective.

A recent paper of a series treated for snoring with an Er:YAG laser showed that the greatest improvement and satisfaction were experienced by patients aged ≥ 50 years [13].

Of the 22 OSAS patients, only 11 patients accepted to test their sleep through a polysomnography after the laser treatment. In those patients who accepted to undergo the polysomnography after the treatment, the measures of AHI before and after the treatment did not show any significant difference ($p = 0.08$). The patients that did not show an AHI improvement at polysomnography nevertheless noticed anyway a subjective improvement and were advised to proceed to further treatments to treat OSAS.

The 40 enrolled patients, after a mean of 20 months of follow-up (14–24), were asked by telephone interview to assess the sustainability of the improvement in snoring reduction. Thirty-two patients agreed to answer to the interview. All the interviews were performed by the same author (IFS) in order not to influence differently the patient. Twenty-three patients (71.9%) affirmed that their improvement was stable, five (15.6%) reported a partial loss of the effect, and only four (12.5%) patients reported loss of effect after 5 to 24 months (median 18.5 months). These latter patients were proposed to perform a new laser treatment if they wished. No patient had reported any subjective side effect at a mean of 20 months of follow-up, even if they did not want to repeat the polysomnography exam.

Discussion

Snoring can be treated in various ways. Lifestyle counseling is very important in patients with sleep disorders and patients should be advised to go to bed having had an early and light supper with no alcohol. Sleeping tablets should be avoided as well as the supine position. Older-generation antihistamine tablets should be replaced in those patients who need them, with new-generation drugs that do not cause sleepiness. Patients should be taught to sleep with the chest slightly up and on one side. A multi-step approach to sleep disorders raising from non-invasive to invasive procedures could be adopted and this could be applied also to snoring and not only to OSAS [23].

With the increase in the quality of life, many patients tend now to refuse a more invasive approach. Also, with greater awareness through the Internet, many know that UPPP and other related surgical procedures may be invasive, painful, and not only fail to improve patient symptoms, but may, in fact, result in a worsening in the patient's condition [24, 25].

Table 3 summarizes the differences in various treatments for snoring.

In this series, all kinds of existing procedures (surgery, MAD, c-PAP in OSAS cases) were illustrated to patients prior to the start of laser treatment and all the patients enrolled in this series refused any kind of invasive or uncomfortable treatment.

The outpatient Er:YAG laser treatments were assessed as non-invasive, non-painful, and had no side effects, even at follow-up, in keeping with other studies [13, 26–28].

Many authors reported that 65–85% of patients responded positively to the Er:YAG laser treatment [13, 26–28]. In this study, the treatment was effective in reducing the loudness of snoring and raising the quality of sleep in many other sleeping disorder symptoms and the results were statistically significant.

In snoring patients, the most frequent site of collapse is the oropharynx (uvula, soft palate, and lateral pharyngeal walls) and the base of the tongue which can be easily reached with the Er:YAG laser (especially the velum). The fact that the collapse can be present at the same time at different section levels (also the epiglottis which cannot be reached with this type of procedure) may explain the lack of results in certain patients. The identification of the site and of the dynamic pattern of obstruction is mandatory in planning the therapeutic decision-making [29] and patients with a relevant “L” collapse (larynx) at the NOHL classification should be excluded from treatment. Another criterion of exclusion for future patients could be the tonsil size obstructing over 50% of the oropharynx, as in these patients, the treatment did not seem to be effective.

A recent study demonstrated that the oropharyngeal airway significantly increased after the treatment as a result of the photothermic effects of the Er:YAG lasers [30]. The shrinkage was also noticed in the histological examination of the soft palate of 20 rats that were exposed to the energy of an Er:YAG laser using the same snoring handpiece (PS04) in a non-contact mode that was used in the present study. A noticeable contraction of the soft palate occurred immediately after laser application and this was also evident histologically in the soft palate after sacrificing the animal [31]. This explains the immediate sensation of a wider throat after the laser treatments that 75% of patients reported in this series and improvement in breathing, as affirmed by the patients. In fact, a lowering of FTP, Mallampati, and the degree of collapse at “O” was observable after the treatment and was statistically significant. Other authors report this improvement of Mallampati after Er:YAG laser treatment for snoring [27].

Some recent articles [26, 30] reported the possibility of Er:YAG laser to be effective also in OSAS patients. In the present study, we observed that it raised the subjective quality of sleep in patients who refused any other kind of treatment, but it did not reduce AHI, so it might not be indicated in OSAS cases. There is the need for further investigation as the number of patients who accepted to perform a polysomnography after treatment is too little. Also, polysomnography was performed after 1 month, but it would have been better to perform it after the neo-collagenogenesis is well established, i.e., after 2–3 months.

One of the limitations of the present study is the lack of an objective measurement of the snoring’s loudness before and after laser treatment. Other authors used the VAS of loudness of snoring to evaluate results after treatments [8].

The results were valued with a patient and his bed partner through interview and it was always performed by the same examiner.

Another limitation of the study is a lack of a control group, as only patients treated with Er:YAG laser were included. A further limitation is that all the patients refused a DISE (drug-induced sleep endoscopy) [32–34] prior to treatment. However, from the results obtained, it can be hypothesized that the nose oropharynx hypopharynx larynx (NOHL) while lying without cushion that was performed with an awake patient was sufficiently reliable.

Conclusions

Er:YAG laser can represent a good alternative to more aggressive standard treatment options for the treatment of snoring, as, comparing to more aggressive surgical and also nonsurgical methods, we report better results for simple snoring with no side effects or risks for the patients. Er:YAG laser treatment is easy to perform and has an extremely high success rate in producing a positive change in sleep patterns. It requires no device to be worn during sleep, involves no chemical treatment and no anesthesia, and does not require a sterile operational field [27].

Nonsurgical and minimally invasive treatment with Er:YAG laser was demonstrated to be effective in a statistically significant way, to reduce the loudness of snoring and many other sleep disorder symptoms by widening the upper airway.

This laser treatment showed no major side effects through long-term follow-up and the results were sustainable at 20 months.

Compliance with ethical standards

Conflict of interest The authors declare that there is no conflict of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

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